

Name: kamanisairam

Reg no:192312195

Ex.No 14: 5G Throughput vs SNR with Modulation Labels

Objective 1:

To evaluate the Throughput (Mbps) of a 5G communication system under different Signal-to-Noise Ratio (SNR) conditions..

Program (Matlab Code)

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation

Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
```



```

title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2 -----

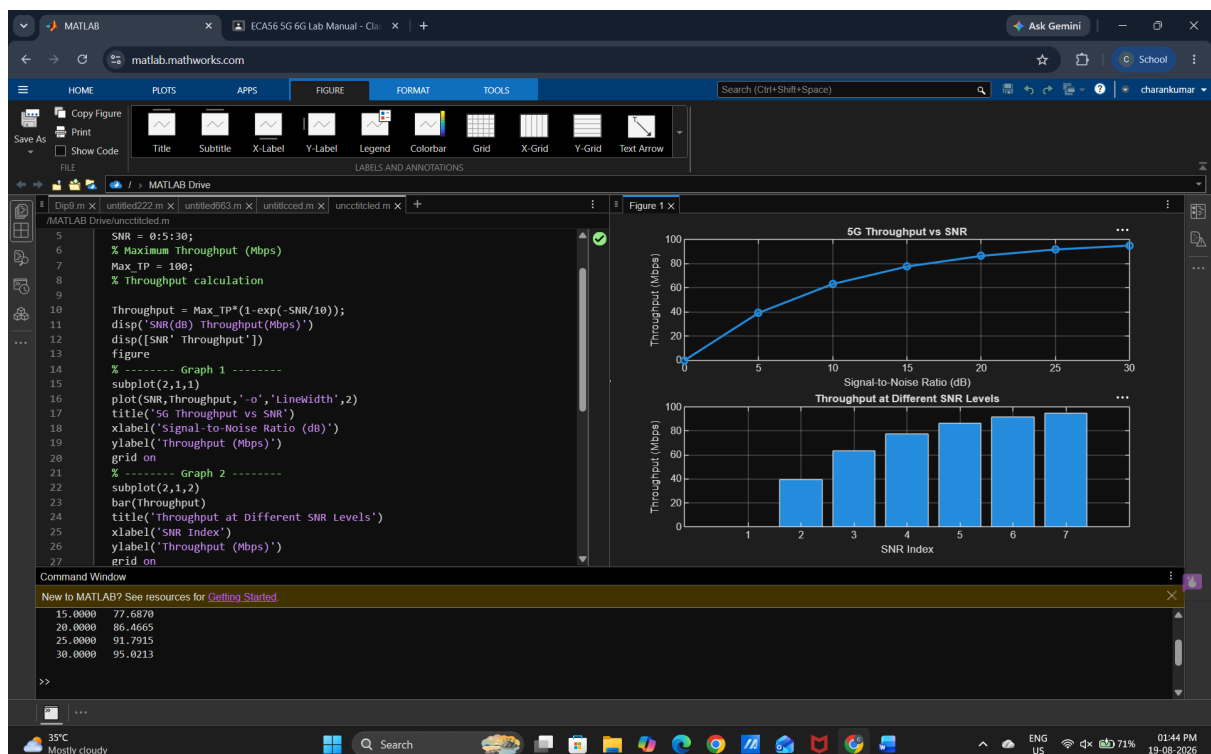
subplot(2,1,2)
bar(Throughput)

title('Throughput at Different SNR Levels')
xlabel('SNR Index')
ylabel('Throughput (Mbps)')

grid on

```

Output:



Objective 2:

To analyze the effect of Signal-to-Noise Ratio (SNR) (dB) on the performance and reliability of the 5G communication system.

Matlab Code

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
% Reliability (%)
Reliability = (1-BER)*100;
disp('SNR(dB) BER Reliability(%)')
disp([SNR' BER' Reliability'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
semilogy(SNR,BER,'-o','LineWidth',2)
title('SNR vs Bit Error Rate in 5G')
xlabel('Signal-to-Noise Ratio (dB)')
```



```
ylabel('Bit Error Rate (BER)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(SNR,Reliability,'-s','LineWidth',2)
```

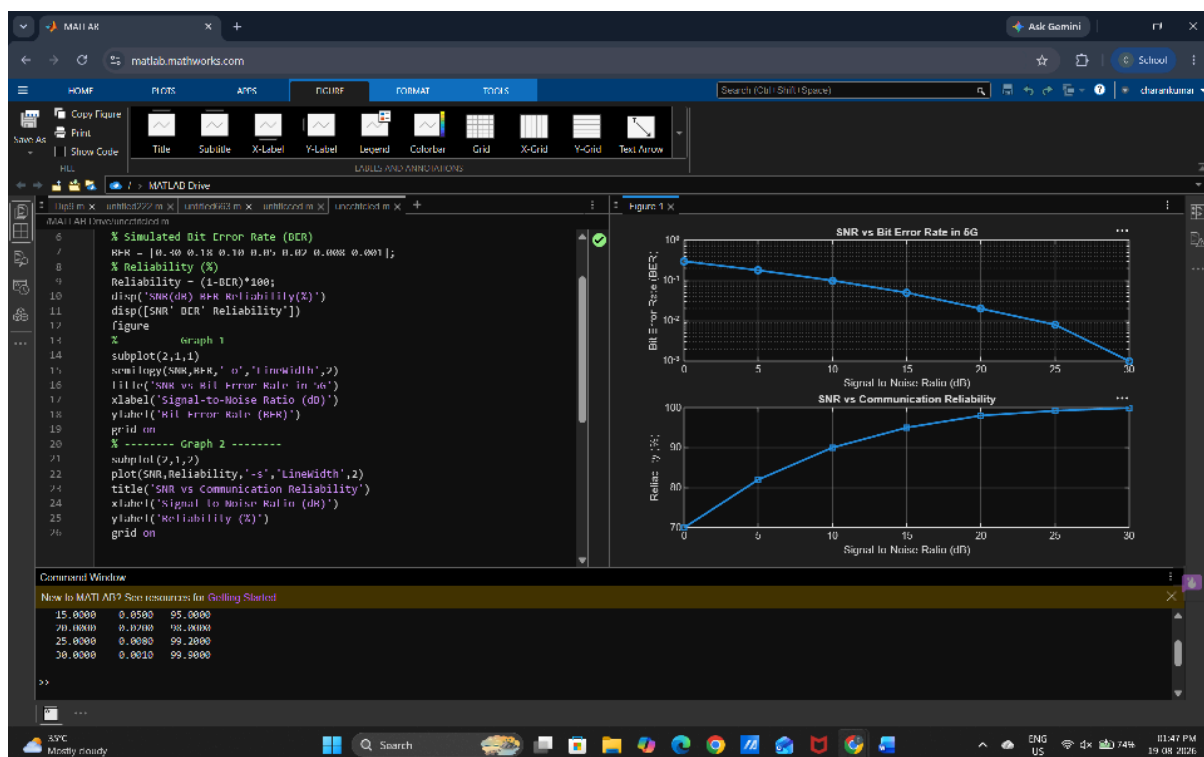
```
title('SNR vs Communication Reliability')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Reliability (%)')
```

```
grid on
```

Output



Objective 3:

To compare the performance of different Modulation Schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM) and determine their suitability for varying SNR conditions..

Matlab Code

```
clc;
clear;
close all;
% Modulation schemes
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
% Required SNR (dB)
SNR = [5 10 18 25];
% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation',SNR',Efficiency', ...
```



```
'VariableNames',{'Modulation','SNR_dB','Efficiency'})))
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,2)
```

```
plot(SNR,Efficiency,'-o','LineWidth',2)
```

```
title('SNR vs Spectral Efficiency')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Spectral Efficiency (bits/s/Hz)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,1)
```

```
plot(1:4,SNR,'-s','LineWidth',2)
```

```
xticks(1:4)
```

```
xticklabels(Modulation)
```

```
title('Required SNR for Different Modulation Schemes')
```

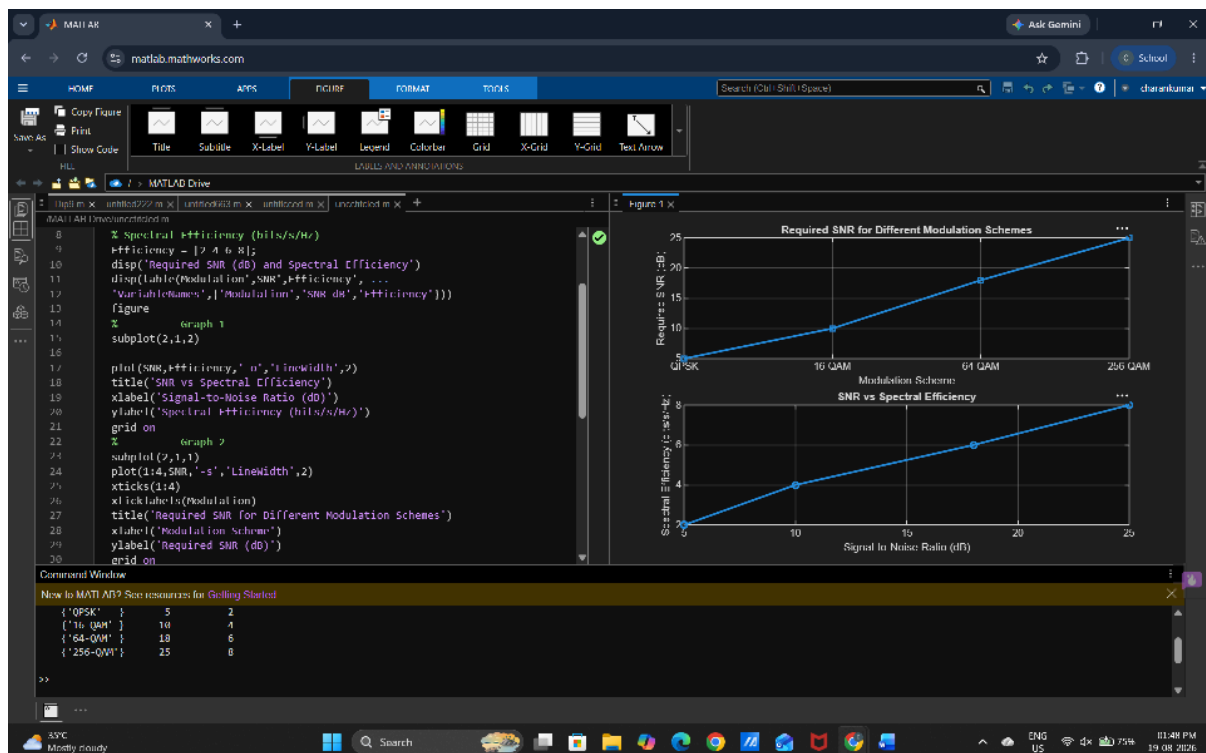
```
xlabel('Modulation Scheme')
```

```
ylabel('Required SNR (dB)')
```

```
grid on
```



Output



Edit with WPS Office